

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

**UCCD1213/UCCG1213/UGCS3213 FUNDAMENTALS OF DIGITAL MEDIA
TECHNOLOGY**
UCCG1113 INTRODUCTION TO MULTIMEDIA

WEDNESDAY, 8 OCTOBER 2025

TIME : 2.00 PM – 4.00 PM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instruction to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A [Total: 75 marks]

This section consists of **THREE (3) COMPULSORY** questions.

- Q1. (a) Define multimedia. (2 marks)
- (b) List **FIVE (5)** elements of multimedia. (5 marks)
- (c) Provide **FIVE (5)** advantages of interactive multimedia. (10 marks)
- (d) Describe **FOUR (4)** principle issues when designing the interface of multimedia applications. (8 marks)
- [Total: 25 marks]
- Q2. (a) Multimedia production involves five main stages: Project Conceptualization, Planning and Costing, Design and Production, Testing, and Delivery.
- (i) Evaluate which **ONE (1)** of these stages consider the most critical for ensuring the success of a multimedia project. (2 marks)
- (ii) Provide **FOUR (4)** clear justification to support your answer in Q2.(a)(i). (8 marks)
- (b) Assume you are comparing two multimedia authoring tools with different media integration capabilities. Identify and explain **FIVE (5)** key criteria you would use to evaluate which tool is more suitable for a multimedia project. (10 marks)
- (c) List **ANY FIVE (5)** roles of multimedia team. (5 marks)
- [Total: 25 marks]

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- Q3. (a) Differentiate between linear and non-linear text in multimedia applications. (4 marks)
- (b) Provide **THREE (3)** advantages of Graphic Interchange Format (GIF). (6 marks)
- (c) Explain **THREE (3)** ways of using digital sound in multimedia project. (9 marks)
- (d) Describe **THREE (3)** basic characteristics of digital video. (6 marks)
- [Total: 25 marks]

Section B [Total: 25 marks]

This section consists of **TWO (2) OPTIONAL** questions. Answer **ANY ONE (1)** question.

- Q4. (a) List **ANY FOUR (4)** areas where multimedia can be used in Education. (4 marks)
- (b) Identify **ANY FOUR (4)** common file formats for audio. (4 marks)
- (c) Explain **THREE (3)** types of animation. (9 marks)
- (d) Virtual Reality (VR) is a computer-generated technology that simulates an interactive and sensory-rich environment. It reacts to the user's movements and inputs, making the virtual experience appear realistic. Identify **FOUR (4)** key differences between non-immersive VR and immersive VR. (8 marks)
- [Total: 25 marks]

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- Q5. (a) Explain **TWO (2)** benefits of using multimedia technology in the following industries: -
- (i) Education. (4 marks)
 - (ii) Business. (4 marks)
- (b) Differentiate between analogue and digital video signals. (6 marks)
- (c) Augmented Reality (AR) blends real-world environments with computer-generated elements, enhancing the user's view by overlaying digital information onto the physical scene. Discuss **ANY THREE (3)** applications of AR. (6 marks)
- (d) Identify **ANY FIVE (5)** common file formats for video. (5 marks)
- [Total: 25 marks]
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